

Probability Contracts: Accuracy, Coherence, and Decisions Across LLM Interfaces

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Abstract

A probability used for a decision should refer to the same event across equivalent requests. We introduce *probability contracts*, a benchmark connecting exact finite-world posteriors, validated event transformations, and failure-aware decision evaluation. Four model-interface configurations are evaluated on 1,000 worlds. Their assessments differ across accuracy, coherence, and decision loss: Kev has lower aggregate canonical posterior error than Jev, but larger complement and coarsening residuals, with accuracy ordering varying by stratum. Jev’s Event and Choice interfaces induce different binary actions on 32.8% of valid pairs at defer cost 0.10. A post-hoc analysis finds that disagreement certifies only 11–52% of mean binary pair error across configurations. An elementary action-region characterization explains when averaging changes decision loss relative to randomly selecting one interface. Although averaging cannot worsen that baseline’s expected Brier score, its decision effect depends on the cost and crossed boundaries; the observed same-baseline penalties occur in configurations already worse than always deferring. The benchmark makes these distinctions measurable without treating consistency as accuracy or a score improvement as a decision guarantee.

1 Introduction

An application can ask whether an event will occur or request a distribution over outcomes and read off the same event’s probability. When the information and event are unchanged, the two reports should agree. Normalization alone does not ensure this agreement, nor does agreement ensure that either report is accurate enough for a decision.

A preserved example from our study makes the distinction concrete. For an event with exact probability $220/317 \approx 0.694$, Jev reports 0.84 through its binary Event interface and 0.97 through a two-option Choice interface. With a wrong answer costing 1 and deferral costing 0.10, the first report leads to deferral; the second leads to answering yes, with expected loss about 0.306. Both outputs are valid. The event is the same,

yet the collected reports lead to different actions. This is a post-hoc illustration; Appendix G gives the requests and selection rule, and Section 6 measures prevalence.

This example raises three questions. *Accuracy*: how close is a report to the correct posterior? *Coherence and stability*: do related reports satisfy probability identities and agree across equivalent representations? *Decision value*: what loss results when an application acts on them? Coherence tests can expose contradictions without knowing the truth, but agreement can also conceal shared error. Conversely, a numerically better forecast need not improve a thresholded decision at a particular operating cost.

We study these questions together using *probability contracts*. A contract specifies the world, revealed information, event partition, representation map, and numerical channel associated with a report. Finite worlds provide exact posteriors, and checked event maps distinguish probability identities and genuine equivalences from changes that remove information. This controlled setting makes posterior error observable rather than inferring it from agreement or from one realized outcome.

Contributions. The benchmark combines exact posteriors, ten validated probability relations, and failure-aware expected decision loss. On the same worlds, it measures errors a consistency check exposes and misses, and connects interface changes to their action-level consequences. A post-hoc analysis quantifies the uncertified error and evaluates fixed reconciliation policies. An elementary action-region characterization relates those empirical policy effects to the controller’s cost thresholds.

The primary study evaluates Kev’s typed decision head, its Qwen3.5-4B base model, the corresponding posttrained model, and hosted Jev-1.13.0. Kev assigns probabilities to supplied options; Jev exposes binary Noul and categorical Choice requests, with Noul called *Event* here. The Qwen baselines use normalized answer-token likelihoods. A smaller extension adds Mistral and OLMo with likelihood and generated-JSON readouts. These comparisons evaluate complete model-interface configurations, not an isolated architectural effect.

2 Related Work

Coherence and uncertainty evaluation. Proper-score dominance provides a classical justification for coherent forecasts (Predd et al., 2009). Zhu and Griffiths (2024) test identities and repeats; Paleka et al. (2025) measure consistency through related forecasts and arbitrage; and Andrews and Sarkar (2026) study richer event relations and irrelevant context. Andrews (2026) develops a general label-free rationality framework, while Wagner et al. (2024) test equivalent span-probability factorizations. Closest in framing, Matta et al. (2026) separate structural coherence, faithfulness, and usefulness. Our distinction among evaluation goals builds on this work. The contribution here is an exact-posterior test bed that measures both the error visible to a consistency check and the error it misses, then evaluates the decisions induced by the same reports.

Known distributions and simulation. Lovering et al. (2025) compare token probabilities with prompt-implied numeric distributions and find identity and order effects. Paruchuri et al. (2024) study probabilistic reasoning about statistical distributions, and Lee et al. (2026) evaluate simulated-world forecasts. We use exhaustive finite enumeration and validated event maps to connect each transformed request to its exact target. This permits direct posterior comparisons for event queries, categorical reports, and their decision consequences.

Calibration and readout. Calibration depends on its target and grouping scheme (Vaicenavicius et al., 2019); grouping loss can persist after apparent calibration (Perez-Lebel et al., 2023). We report calibration alongside per-world error and fit no test-set recalibration (Guo et al., 2017). Readout and belief-choice sensitivity motivate specifying the numerical channel separately from the decision rule (Mahaut et al., 2024; Kim and Kang, 2026; Yamin et al., 2026; Kudum et al., 2026). Label preferences are a possible readout confound (Zhao et al., 2021b); a large complement residual alone does not identify their cause. Appendix N covers additional concurrent Jev studies.

Scoring rules and decisions. Proper scoring rules and threshold decisions have a classical relationship (Schervish, 1989; Gneiting and Raftery, 2007). Mixture representations motivate Murphy diagrams, which compare elementary scores across thresholds (Ehm et al., 2016). Our controller uses the reject option of Chow (1970); decision calibration likewise ties calibration requirements to downstream actions (Zhao et al., 2021a). We use this theory to interpret measured losses, not claim a new score-decision separation. Concurrently,

Zhang et al. (2026) show that similar aggregate scores on ContractNLI can conceal different example-level decisions across request conditions. Here, exact event posteriors additionally identify probability error and expected decision regret on each world.

3 Probability Channels and Their Targets

LLM interfaces expose different objects under the name *probability*. A vocabulary softmax describes the next token given a prefix. Normalizing selected answer-token probabilities conditions further on that answer set and discards mass assigned elsewhere. Sequence likelihood concerns an exact string, including its wording and tokenization. Generated percentages are numerical claims expressed as text. None of these constructions alone establishes a posterior over an application’s semantic events.

We call an output a *declared-event probability* when the interface assigns normalized mass to an exhaustive event set in the request. Jev’s Choice and Noul interfaces declare this target (TypeSafe AI, 2026); Kev’s head maps to supplied options (Kev contributors, 2026c). Our likelihood readouts are conditional proxies over answer IDs. A separate confidence field is not treated as an event probability unless its documented semantics support that interpretation.

Validation first checks support, finite nonnegative values, and total mass within tolerance. Scoring then asks whether a valid report is accurate, coherent with related reports, and useful for the specified controller. Keeping these stages separate lets us compare numerical channels without equating their underlying meanings.

4 Probability Contracts

4.1 Measurement object

A probability contract records a finite world, its revealed information, an output partition, a representation map, a numerical channel, and the retained requests and responses. A *root* is one generated world and information set together with its related requests; it is the unit of paired evaluation. Equivalence requires a checked map preserving the queried event and information. Relations between different events, such as complementation or conditioning, instead require the corresponding probability identity.

Each world has four latent cells indexed by Boolean attributes a, b . Cell i has integer prior count w_i , rational evidence likelihood ℓ_i , and a category label. Its posterior mass is

$$q_i = \frac{w_i \ell_i}{\sum_j w_j \ell_j}.$$

An event’s probability sums the masses of its cells; category probabilities likewise sum cells with the same label. The finite-population generator filters counted objects by observed b ; noisy-sensor applies a rational sensor likelihood; joint-attributes samples from counted Boolean cells; and missing-rule maps cells to three observable categories. The first three have four-category canonical partitions and the last has three.

The primary oracle uses exact rational arithmetic. A separately implemented path normalizes the prior in 70-digit decimal arithmetic before the likelihood update; admission requires agreement within 10^{-15} . A typed event algebra checks atoms, set membership, negation, conjunction, disjunction, and conditioning. Partitions must be disjoint and exhaustive on positive posterior support, and conditioning events must have positive mass.

4.2 Representations and relations

Each root generates a canonical exhaustive distribution and eleven variants. Event A contains the first $\lfloor K/2 \rfloor$ sorted categories; $\neg A$ is its complement. Event B uses attribute b , with further requests for $A \cap B$ and $A \mid B$. Binary Choice presents the same event as Event, and an exact repeat duplicates the Event request. The remaining variants permute options, change layout within an encoding family, bijectively rename provider keys, or append independently assigned metadata.

Table 1 specifies ten relation-level tests. Complement, coarsening, intersection, and product assess compatibility among event claims. Interface, permutation, rendering, key, and metadata comparisons assess stability after semantic alignment. Exact repeat measures variability under an identical request, including service nondeterminism. Their residuals retain their own meanings and are not combined into one coherence score.

Each transformation records its world, information set, target, and provider-to-canonical event map. Vector comparisons invert that map before scoring; scalar comparisons use the event algebra. Undefined conditionals are recorded as not applicable and are never sent. Product comparisons are eligible on 913 of 1,000 roots; all other variants are eligible on every root. Continuous residuals measure discrepancy size, while prespecified tolerances summarize pass rates.

5 Experimental Design

5.1 Worlds and configurations

The primary cohort contains 1,000 roots: 250 per generator and 125 per generator–encoding stratum, using joined tables or nested records. Semantic hashes separate it from 2,700 earlier benchmark roots; they do not establish absence from pretraining data. Twelve

requests per root give 12,000 intended slots per configuration. Excluding 87 undefined conditionals leaves 11,913 eligible requests.

The local configurations are Kev-4B (checkpoint 485ace8...), Qwen3.5-4B-Base (1001bb4...), and posttrained Qwen3.5-4B (851bf6e...) (Kev contributors, 2026a; Qwen Team, 2026a,b). Kev adds a rank-16 LoRA adapter and a 256-dimensional pointer head (Kev contributors, 2026c). The pinned source release card describes public classification and generated policy/rule training, with temperature fitted on in-distribution development rows (Kev contributors, 2026b). We retain the stored $T = 2.143546925$; none is fitted for this study. Appendix A.1 separates that documented provenance from unverified training overlap. The Qwen readouts normalize answer-ID logits over the specified IDs while retaining their full-vocabulary mass. Posttrained uses its non-thinking template. All three run in FP32; Kev’s rounded output is a separate sensitivity. Base (also *matched base* in archival tables) denotes the Qwen base likelihood configuration.

Jev-1.13.0 is queried through its native Noul and Choice interfaces. A separate 477-request development allocation checks technical compatibility and is excluded from estimates. Prompts, thresholds, and the main cohort were fixed before collection. Each main request has one recorded attempt. Jev returns serialized decimal probabilities, making numerical precision part of the evaluated configuration.

A supplementary 200-world check adds Mistral and OLMo with likelihood and unconstrained generated-JSON readouts. It tests additional configurations, not a new cohort or a second native-interface implementation (Appendix M).

5.2 Validity and scoring populations

Collection cannot access evaluator posteriors. Exact requests and raw responses are recorded before validation, including rejected responses. Each eligible request is valid, rejected after a response, or unavailable. Probability metrics use valid outputs; paired contrasts require valid constituents on the same root. Decision evaluation instead retains every intended root and assigns rejected or unavailable outputs to defer. This evaluates the full pipeline, but low-cost fallback can also lower loss by reducing coverage.

Native validity is the valid fraction of eligible requests. Service availability additionally counts rejected HTTP-200 responses as served. Bounded identical-payload retries are permitted only for designated transient failures; no main Jev request used that allowance (Appendix D).

The operational criterion fixed before Jev collection requires native validity at least 0.995 and a lower 95% root-bootstrap pass-rate bound at least 0.95 for each of the nine primary relations in Table 1, excluding its

Table 1: Declared probability relations. p_X denotes the output after canonical event alignment. Continuous residuals are primary; pass thresholds are prespecified summaries.

Relation	Aligned claims	Residual or condition	Eligibility
Complement	$p(A), p(\neg A)$	$ p(A) + p(\neg A) - 1 $	Exhaustive binary event
Coarsening	$p(A), p(\omega)$	$ p(A) - \sum_{\omega \in A} p(\omega) $	Same partition and information
Event/Choice	$p_{\text{event}}(A), p_{\text{choice}}(A)$	Absolute difference	Bijection to same event
Intersection	$p(A), p(B), p(A \cap B)$	Distance outside Fréchet bounds	Aligned A, B
Product	$p(A \cap B), p(B), p(A B)$	$ p(A \cap B) - p(B)p(A B) $	$q(B) > 0$
Permutation	$p(\omega), p_{\pi}(\omega)$	Total variation after inverse map	Bijection; same information
Rendering	$p(\omega), p_r(\omega)$	Total variation	Same encoding family
Opaque keys	$p(\omega), p_k(\omega)$	Total variation after key map	Bijection
Irrelevant metadata	$p(\omega), p_m(\omega)$	Total variation	Metadata independent by design
Exact repeat	$p(A), p'(A)$	Absolute difference	Identical payload

intersection-bounds diagnostic. Repeat uses tolerance 0.02; the others use 0.05. Continuous residuals accompany this threshold-based criterion.

5.3 Accuracy, decisions, and uncertainty

Posterior error is primarily $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$. Expected multiclass Brier is $\sum_y q_y \sum_k (p_k - \mathbf{1}[k = y])^2$, and expected native log loss is $-\sum_y q_y \log p_y$. Positive q_y with $p_y = 0$ gives infinite log loss, which remains in the result. Realized scores use one outcome draw per root. Prespecified calibration compares top-class confidence with these draws using fixed 10-, 15-, and 20-bin schemes and an equal-frequency sensitivity. A separately labeled post-hoc summary instead uses the exact posterior mass of the selected class. Both describe grouped reliability; TV measures per-root error. No recalibration is fitted.

The reject-option controller (Chow, 1970) chooses a class or defers. Wrong and correct answers cost 1 and 0; deferral costs c . The primary cost is 0.10, with 0.02, 0.05, 0.20, and 0.40 also fixed before collection. The controller minimizes expected cost under the report. For a normalized vector it acts when $\max_i p_i \geq 1 - c$, otherwise deferring; ties prefer the first canonical class before defer. Expected loss evaluates the chosen action under q , and oracle regret subtracts the minimum loss achievable under q . Always defer is the cost- c reference; a gold-informed best fixed action is descriptive only.

Canonical multiclass decisions and binary event decisions are scored separately. For a binary report $p(A)$, the controller uses $(1 - p(A), p(A))$. Action-change rates compare aligned binary Event/Choice requests on paired-valid roots. They are reported beside deferral because two always-defer policies can agree in action despite different probabilities.

Aggregates weight the eight generator-encoding strata equally. Valid-only estimates average their nonempty strata and disclose omissions. We use 5,000

paired root-bootstrap replicates (seed 2026092408), re-sampling within strata with all related requests and configurations together. Each statistic retains its own validity denominator. Intervals are pointwise percentile summaries, not multiplicity-adjusted tests or equivalence bounds; they quantify variation over worlds, not over training runs or repeated service collection.

6 Primary Study: 1,000 Worlds

Figure 1 compares posterior error, interface disagreement, and decision loss. Each answers a different evaluation question.

6.1 Posterior accuracy

Kev has the lowest aggregate canonical TV: 0.2350, compared with 0.3740 for base, 0.3202 for posttrained, and 0.2677 for Jev (Appendix Table 6). The first three estimates use 1,000 valid roots; Jev uses 990. Paired differences favor Kev over base by -0.1390 $[-0.1494, -0.1286]$ and over posttrained by -0.0853 $[-0.0942, -0.0761]$. Posttrained minus base is -0.0537 $[-0.0608, -0.0465]$.

On 990 paired-valid roots, Jev minus Kev is 0.0335 $[0.0222, 0.0448]$. Jev minus base is -0.1065 $[-0.1190, -0.0940]$, and Jev minus posttrained is -0.0524 $[-0.0633, -0.0413]$. These paired differences need not equal differences of the full marginal means because validity changes the comparison population.

Kev and Jev each have lower canonical TV in four of the eight strata (Appendix Table 14); one Jev advantage is near a tie, 0.1858 versus 0.1866. Kev’s aggregate advantage is concentrated in joint-attributes and noisy-sensor strata. On missing-rule joined tables, its TV is 0.4545 versus Jev’s 0.1931. Thus the reported macro-average is not uniform dominance. Strata contain different roots; within-root rendering tests, not these contrasts, assess representation stability.

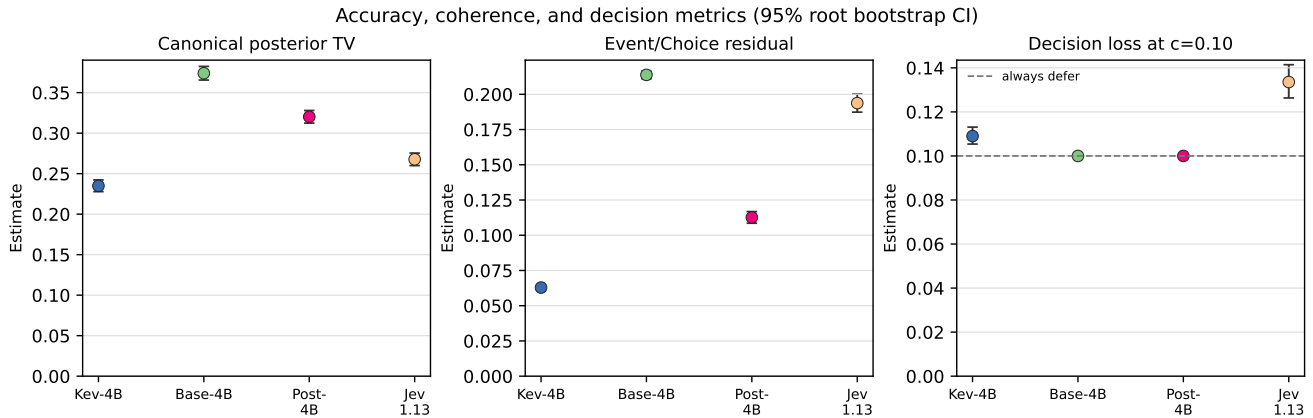


Figure 1: Canonical posterior error (left), binary Event/Choice disagreement (middle), and canonical decision loss (right); lower is better. Error bars are 95% stratified root-bootstrap intervals, sometimes narrower than the markers. Jev TV uses 990 valid canonical roots; the other plotted estimates use 1,000 roots. The outer panels use the canonical multiclass task; the middle panel compares two binary reports of the same event.

Expected log loss is finite for all original local canonical outputs. Jev has 20 infinite expected and 5 infinite realized log losses among its 990 valid roots. The post-hoc 15-bin exact-posterior calibration gaps are 0.0618, 0.1441, 0.0877, and 0.1952 for Kev, base, posttrained, and Jev. They compare confidence with the posterior mass of the selected class, avoiding outcome-draw noise without identifying per-root accuracy. Appendix I retains the distinct prespecified sampled-outcome summaries.

6.2 Coherence and representation stability

Kev’s lower aggregate canonical error does not make it more coherent on every relation (Table 2). Its complement residual is 0.5055 versus Jev’s 0.0929; coarsening residuals are 0.3554 and 0.1751. Conversely, Jev’s Event/Choice gap exceeds Kev’s and posttrained’s. Posttrained has the largest complement residual, 0.8000, and Kev the largest intersection-bound residual, 0.1553. A post-hoc signed-sum check shows excess mass for every Kev and posttrained complement pair (Appendix L); this describes the failure without attributing it to label bias, negation, or arithmetic.

Exact repeats provide a distinct control. Local deterministic repeats agree exactly, whereas Jev’s mean repeat residual is 0.0188. Although this mean lies below the 0.02 tolerance, only 745/1,000 pairs pass: 0.7450 [0.7180, 0.7720]. A mean below tolerance therefore does not imply a high pairwise pass rate. Repeat variability includes the served system and cannot be attributed solely to the model.

Jev fails the operational criterion: validity is below 0.995, and all nine primary relation pass-rate lower bounds are below 0.95. Applying the same criterion to

the local configurations also gives no pass; this symmetric extension is post-hoc (Appendix I). The relation-level magnitudes, rather than the conjunction alone, show how the failures differ. Appendix J includes rendering, opaque keys, and irrelevant metadata, with eligibility, support, and intervals.

6.3 Decision consequences

Table 3 separates two decision problems. Its first three metric columns evaluate the canonical multiclass report; its final column compares two binary reports of event A . This distinction explains how posttrained can always defer on the canonical task yet change actions between binary Event and Choice requests.

For binary Event/Choice at $c = 0.10$, actions change on 14.9% of roots for Kev, 34.2% for posttrained, and 32.8% for Jev. Base always defers and has no action changes despite its probability gap. All 328 Jev changes are act/defer crossings. At $c = 0.40$, 62 of 405 changes instead switch class labels. Probability disagreement becomes consequential when it crosses a boundary of the specified controller.

On the canonical task at $c = 0.10$, oracle loss is 0.0897: only 0.0103 average improvement is available below always defer, with positive headroom on 119/1,000 roots. Kev defers on 96.1% of roots yet incurs loss 0.1090, exceeding always defer by 0.0090. Jev’s loss is 0.1336; base and posttrained equal always defer. At $c = 0.40$, Kev’s loss is 0.3906, now 0.0094 below always defer. The full cost grid (Figure 2) makes this operating-point dependence explicit.

For this loss, posterior TV δ bounds oracle regret by 2δ , or by δ when the predicted and oracle actions differ only between acting and deferring. The relevant quantity is error against truth, not merely agreement

Table 2: Mean relation residuals (lower is better within a row). Jev uses 1,000 valid pairs for Event/Choice, complement, and repeat; 990 for coarsening; 999 for intersection; 912 for product; and 976 for permutation. Local support is complete (913 eligible product roots, otherwise 1,000). All intervals and relations are in Appendix J.

Relation	Kev	Base	Posttrained	Jev
Event/Choice	0.0629	0.2138	0.1126	0.1938
Complement	0.5055	0.1155	0.8000	0.0929
Coarsening	0.3554	0.2451	0.2836	0.1751
Intersection bound	0.1553	0.0814	0.0358	0.0200
Product identity	0.1328	0.2682	0.1406	0.0836
Permutation TV	0.1143	0.3224	0.3470	0.0534
Exact repeat	0.0000	0.0000	0.0000	0.0188

Table 3: Decisions at defer cost 0.10: means and 95% root-bootstrap intervals. The first three metrics use the canonical multiclass task; the final column compares separate binary Event/Choice requests. Each uses 1,000 roots per configuration. Canonical loss includes failure-triggered deferral. Always-defer loss is 0.10.

Configuration	Canonical multiclass task			Binary task
	Expected loss	Oracle regret	Defer rate	Event/Choice changes
Kev typed	0.1090	0.0193	0.9610	0.1490
	[0.1054, 0.1131]	[0.0158, 0.0232]	[0.9500, 0.9720]	[0.1290, 0.1710]
Base likelihood	0.1000	0.0103	1.0000	0.0000
	[0.1000, 0.1000]	[0.0088, 0.0118]	[1.0000, 1.0000]	[0.0000, 0.0000]
Posttrained likelihood	0.1000	0.0103	1.0000	0.3420
	[0.1000, 0.1000]	[0.0088, 0.0118]	[1.0000, 1.0000]	[0.3210, 0.3630]
Jev-1.13.0	0.1336	0.0438	0.8160	0.3280
	[0.1263, 0.1414]	[0.0366, 0.0515]	[0.7920, 0.8380]	[0.3020, 0.3540]

between reports. Appendix H gives the elementary bounds and the margins needed to preserve an action across representations.

7 What Disagreement Misses and Averaging Changes

We use the preserved binary reports to ask two follow-up questions: how much posterior error can disagreement certify, and does enforcing agreement improve decisions? This analysis is *post hoc*: its specification was recorded after inspecting descriptive certificate ratios. All 1,000 Event/Choice pairs and their repeats are valid for each configuration. The policies below use only reported probabilities, without fitting or access to evaluator truth.

This section evaluates binary event A , not the canonical multiclass outcome. At $c = 0.10$, binary oracle loss is 0.08132, leaving 0.01868 available below always defer, with positive headroom on 208 roots.

7.1 Disagreement certifies only part of error

Let p, p' report the same event with exact probability q , and let $m = (p + p')/2$. Two standard identities give

$$\frac{|p - q| + |p' - q|}{2} = \frac{|p - p'|}{2} + d(q, [p \wedge p', p \vee p']), \quad (1)$$

$$\frac{(p - q)^2 + (p' - q)^2}{2} = \frac{(p - p')^2}{4} + (m - q)^2. \quad (2)$$

Here d is distance to the interval between the reports. Half the absolute gap is an observable lower bound on mean pair error. The remaining error requires knowledge of q : reports can agree while both miss the posterior. The squared identity similarly separates observable disagreement from the midpoint’s error. We use these identities as diagnostics, not as a new coherence theorem.

For Jev, the absolute-error certificate accounts for 0.522 of mean pair error; for Kev, 0.107. The corresponding fractions are 0.416 for base and 0.149 for posttrained. Squared-error fractions are 0.232, 0.014, 0.131, and 0.031 in the same order. These are ratios of mean components, not averages of per-root ratios. Kev’s smaller interface gap therefore certifies a smaller share of its binary error, despite its lower canonical TV.

On these same binary pairs, Jev also has lower mean absolute error than Kev (0.18567 versus 0.29397), despite its larger Event/Choice gap (0.1938 versus 0.0629). The accuracy-coherence ordering thus reverses even on a common target and support, not only between canonical and binary tasks. Appendix O gives absolute components, intervals, and complement checks.

7.2 A better Brier score need not mean a better decision

Averaging is the equal-weight Euclidean projection of (p, p') onto equality. By Equation 2, replacing the pair with m reduces its average expected one-coordinate Bernoulli Brier score by $(p - p')^2/4 \geq 0$. The outcome-variance term $q(1 - q)$ cancels; the two-coordinate categorical binary convention doubles all terms. This standard score guarantee (Predd et al., 2009) is relative to the *average* report, not each individual report, and does not concern thresholded decision loss.

Proposition 1 (action-region characterization).

For $0 < c < 1/2$, let $a_c(x)$ be no for $x \leq c$, defer for $c < x < 1 - c$, and yes for $x \geq 1 - c$. Their true losses are $q, c, 1 - q$. Writing $\ell_c(x; q) = L_c(a_c(x); q)$, the averaging penalty is

$$\Delta_c = \ell_c(m; q) - \frac{1}{2}\{\ell_c(p; q) + \ell_c(p'; q)\}. \quad (3)$$

It is zero if both reports select the same action. For adjacent action regions it equals half the selected-minus-unselected true action loss. For opposite class actions it is $q - 1/2$, $c - 1/2$, or $1/2 - q$, according as the midpoint selects no, defer, or yes. In particular, resolving opposite class actions to defer always helps. The proof is a case analysis on convex action regions (Appendix P). Thus gap size alone cannot determine the sign: the cost, crossed regions, and true posterior matter.

We compare Event-only and Choice-only with two fixed two-query policies: use the mean report, or take the common action and otherwise defer (*agreement-deferral*). Event plus its exact repeat supplies equal-query controls for both rules. Missing constituents would trigger defer. The average-single-interface comparator chooses Event or Choice uniformly, independently of the world. It requires one query; the reconciliation policies require two.

At the primary cost $c = 0.10$ (Table 4), averaging reduces loss relative to that randomized comparator by 0.03463 for Jev, 0.00959 for Kev, and 0.05297 for posttrained. Base stays at always defer. Kev and posttrained remain worse than always defer under both two-interface policies.

Improvement over the randomized interface need not beat a fixed individual interface. For Jev at $c = 0.10$, averaging increases loss over Event by 0.00236 [0.00077,

0.00411], while coverage rises from 0.062 to 0.104. Both losses remain below always defer, but the Brier guarantee is not relative to Event alone. Agreement-deferral equals Event loss; the Event/repeat mean has loss 0.09645 at 0.057 coverage.

Against the same randomized baseline used by the Brier guarantee, averaging has higher mean loss in 3 of the 20 configuration-cost cells: Kev at $c = 0.20$, and base and posttrained at $c = 0.40$ (Appendix Table 26). In all three, both policies are worse than always defer. Posttrained’s penalty is 0.00347 [0.00122, 0.00581]. These post-hoc results illustrate the score-decision distinction, but do not demonstrate a useful high-accuracy deployment setting. Appendix P gives a separate analytic example with arbitrarily accurate reports and both policies better than defer; it is not model evidence.

8 Discussion and Limitations

The benchmark’s main lesson is about the unit of evaluation: a probability report belongs to an event, information set, numerical channel, and decision rule. Canonical posterior accuracy does not determine coherence on separate binary queries. Agreement between reports leaves shared error unchecked. Even reconciliation with a proper-score guarantee can change threshold crossings in a way that increases the chosen loss. These distinctions matter when deciding which interface to trust for a particular action, rather than seeking one overall model ranking.

Exact finite worlds trade breadth for identifiable targets. Four latent cells and two encodings allow checked event alignment and exact posterior evaluation, but do not establish performance on open-ended factual or deployment tasks. Ordinary correctness labels would not, by themselves, reveal those tasks’ latent posteriors. Transfer to routing, retrieval, or other applications remains an external-validity question.

The main local models share a Qwen lineage, and the hosted study covers one Jev revision. The two-model 7B extension has only 59 and 60 valid canonical JSON reports out of 200; its Event/Choice prompts are identical controls, not replications of the native contrast. Training, prompts, output mechanism, and precision vary together. Large posterior errors and low answer-token mass limit conclusions about accurate reasoners; neither a typed-head effect nor a particular readout-failure mechanism is identified. Small oracle headroom at the primary cost also limits possible decision gains.

The symmetric criterion, exact-posterior calibration, decision-boundary decomposition, rounding sensitivity, collection-window comparisons, and certificate/policy study are post-hoc. Signed complement summaries and the action-region presentation were added during revision. Pointwise intervals do not cover unrestricted

Table 4: Post-hoc binary policy loss at $c = 0.10$ (all 1,000 roots per configuration; lower is better). Parentheses in headers give requests per decision. Agreement means common action or defer. Always-defer loss is 0.10000 and binary oracle loss is 0.08132. Intervals and full comparisons are in Appendix O.

Configuration	Event (1)	Choice (1)	Mean (2)	Agreement (2)	Repeat mean (2)
Kev	0.16695	0.15786	0.15281	0.14186	0.16695
Base	0.10000	0.10000	0.10000	0.10000	0.10000
Posttrained	0.25448	0.23257	0.19055	0.17239	0.25448
Jev	0.09654	0.17052	0.09890	0.09654	0.09645

metric searches; five costs do not establish dominance throughout a continuum. Historical service order, omitted monetary query costs, and loss-reducing failure deferral further limit policy comparisons. We therefore report losses alongside coverage, query counts, and the fallback baseline.

9 Conclusion

Probability contracts connect exact posterior accuracy, relation-specific coherence, and decision loss in one controlled evaluation. Their value is to quantify distinctions that consistency or accuracy alone cannot resolve: shared error, interface-induced action changes, and cost-dependent reconciliation effects. The empirical penalties and weak configurations delimit the findings; the elementary characterization explains their mechanism without turning them into a universal ranking. A probability interface should be evaluated with its event semantics, coverage, and decision cost specified together.

AI Use Statement

OpenAI ChatGPT and Codex assisted with research planning, benchmark and transformation design, implementation, tests, experiment orchestration, analysis, literature discovery, artifact preparation, and manuscript drafting and editing. Anthropic Claude and Google Gemini supplied manuscript critiques that informed the revision. The finite-world datasets were generated deterministically by code, rather than sampled as model-written examples. Evaluated outputs were preserved as collected; writing assistants did not alter, impute, or fabricate predictions or reference posteriors.

Verification includes two oracle calculations, mathematical and implementation tests, independent result checks, source checks for citations, and offline replay. These checks establish specific computational properties; they do not substitute for author judgment about the interpretation or prose. The authors are responsible for the final content, including all AI-assisted material. Generative AI systems are not authors.

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A Reproducibility Checklist

- Models.** Local model, source, tokenizer, and run-time revisions are pinned. The served model is Jev-1.13.0 and its response fingerprint is validated.
- Data.** Deterministic generators, semantic hashes, seeds, exact event maps, partitions, and eligibility records are retained. The primary cohort has 1,000 roots and 11,913 eligible requests per configuration; the extension reuses 200 roots with 2,385 requests per new readout.
- Gold isolation.** Request construction and collection cannot access evaluator gold; regression tests check the boundary.
- Metrics.** Primary TV, Brier, native log loss, calibration, relation residuals, risk-coverage, regret, deferral, and action changes were fixed before Jev outcomes. Certificate decomposition and derived-policy comparisons are explicitly post-hoc and have a separately preserved specification.

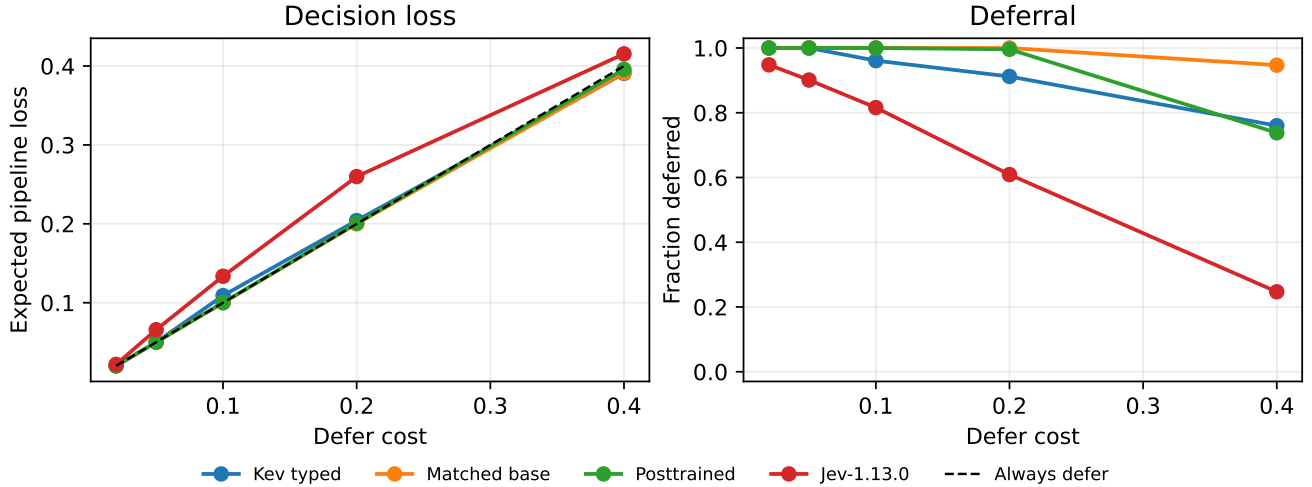


Figure 2: Post-hoc visualization of the five prespecified costs (connecting lines guide the eye, not an evaluation of intervening costs). Expected decision loss and deferral are shown separately; the dashed loss line is always defer. The primary cost is 0.10. Discussed in Section 6.

5. **Uncertainty.** 5,000 paired root bootstrap replicates preserve transformations and use eight equal-weight strata.
6. **Failures.** Raw responses precede validation. Invalid HTTP-200 responses are rejected without repair or retry. Only prespecified transport failures permit exact-payload retry.
7. **Audit.** An independent path checks request bytes, raw bytes, attempt/ledger conservation, retry rules, version, budgets, terminals, and scoring boundary.
8. **Compute.** The original local configurations used FP32; the two-model extension used BF16. Jev collection requires no GPU. Hardware, runtimes, commands, and receipts accompany the artifact.
9. **Artifacts.** Credential-free code, manifests, tests, analysis, and replay material are separated from credentials, weights, caches, and raw archives.

A.1 Documented Kev training provenance

The release card at pinned source revision 557598f describes the `decision-v7` suite: 10,000 public classification records, 896 policy minimal-pair records, and 1,680 generated-rule records. The LoRA adapter and newly initialized pointer head were trained on option-distribution cross-entropy. A later update mixed 1,425 generated date/evidence-removal records with 2,000 replay records. The stored temperature was fitted by negative log-likelihood on in-distribution development rows (Kev contributors, 2026b). These are release-author descriptions; the compact artifact does not independently

bind the training manifests to the evaluated checkpoint. The benchmark’s semantic-hash separation from its own 2,700 earlier roots is not a training-data overlap audit. We have not established whether the released training suites contain benchmark-equivalent worlds.

B Full Variant Inventory

Table 5 lists each request slot. The undefined conditional slots remain in the inventory but are excluded before collection.

Table 5: Per-configuration main requests.

Variant	Intended	Eligible
Canonical	1,000	1,000
Event A	1,000	1,000
Complement $\neg A$	1,000	1,000
Event B	1,000	1,000
Intersection	1,000	1,000
Conditional	1,000	913
Binary Choice	1,000	1,000
Permutation	1,000	1,000
Rendering	1,000	1,000
Opaque keys	1,000	1,000
Irrelevant metadata	1,000	1,000
Exact repeat	1,000	1,000
Total	12,000	11,913

Table 6: Canonical posterior accuracy: eight-stratum means with 95% paired root-bootstrap intervals. Scores use valid canonical roots; all 1,000 intended roots enter decision loss. Full response counts and realized Brier scores appear in Appendix C.

Configuration	Valid canonical roots	Posterior TV	Expected Brier
Kev typed	1,000	0.2350 [0.2278, 0.2423]	0.6333 [0.6229, 0.6436]
Base likelihood	1,000	0.3740 [0.3655, 0.3824]	0.7474 [0.7379, 0.7571]
Posttrained likelihood	1,000	0.3202 [0.3124, 0.3280]	0.6938 [0.6831, 0.7048]
Jev-1.13.0	990	0.2677 [0.2599, 0.2754]	0.6560 [0.6415, 0.6705]

C Additional Accuracy and Accounting

Each configuration has 12,000 intended slots and 11,913 eligible requests; 87 conditionals are undefined. All three local configurations returned valid vectors on every eligible request. Jev has 11,849 valid, 62 rejected, and 2 unavailable responses. Table 7 gives the corresponding realized canonical Brier scores.

Table 7: Realized canonical Brier: mean [95% interval].

Configuration	Realized Brier
Kev typed	0.6359 [0.6160, 0.6571]
Base likelihood	0.7530 [0.7332, 0.7739]
Posttrained likelihood	0.6918 [0.6696, 0.7137]
Jev-1.13.0	0.6469 [0.6094, 0.6851]

D Collection Accounting

Every main Jev request has one recorded transport attempt: 11,849 valid, 62 rejected after HTTP 200, and two unavailable after HTTP 403 and HTTP 520 respectively. These two statuses were ineligible for main retries. The protocol allowed at most six identical-payload attempts for HTTP 429, HTTP 529, or pre-response network failures, but no main request used that allowance. Two later one-shot service-restoration probes returned HTTP 200; their outputs are excluded and the original unavailable outcomes remain unchanged.

Main results required complete request accounting, an independent audit, version and exact-payload verification, relation-specific valid denominators, and consistency between the machine-readable results and manuscript. The 477-request development cohort is excluded from all estimates. The compact submission supplement contains selected scores, analysis products, code, and audit records. Complete raw-response

archives are retained separately; the compact artifact alone does not replay the full collection from raw bytes.

E Decimal Boundary Handling

Pass classification uses the decimal values serialized in the responses, with inclusive comparisons to the specified tolerance. Binary floating-point subtraction can represent an exact decimal difference of 0.02 as slightly larger than 0.02; treating that as a failure would alter the intended rule. The corrected calculation and an independent implementation agree that 745 of 1,000 Jev exact-repeat pairs pass. This numerical correction changes neither responses nor tolerances. Earlier analyses are retained for traceability. All current counts use the decimal-domain comparison, and the joint criterion remains unmet. The extension’s strict JSON parser likewise uses an inclusive decimal simplex tolerance of 10^{-6} . One accepted vector (0.333333, 0.333333, 0.333333) lies exactly on this boundary. The offline validator sums the stored decimal values exactly so binary roundoff does not reject it; probabilities and collection statuses remain unchanged. Canonical expected-cost calculations use the supplied vector without normalization. Relation-specific binary controllers use the reported scalar $p(A)$ and its logical complement, distinct from scoring the full canonical vector.

F Collection Continuity Diagnostic

Immutable receipts partitioned the main run into 174 pre-HTTP403 requests (172/1/1 valid/rejected/unavailable), 6,495 post-HTTP403 and pre-HTTP520 requests (6,452/42/1), and 5,244 post-HTTP520 requests (5,225/19/0). No admitted request was removed. Fixed-stratum, fixed-primitive differences were mixed around the 403. After the 520, the two supported Choice cells had higher TV, Brier,

and regret, while Noul changes were small and mixed. The request order changed roots and generator support across windows, so this is a descriptive continuity check rather than a causal temporal comparison.

G An Action-Changing World

For a post-hoc illustration, we select a root by a fixed hash ordering among pairs where Jev’s event answer strictly defers and its Choice answer strictly acts. This illustrates a mechanism, not its prevalence. The selected noisy-sensor world has prior counts (67, 85, 9, 79) and sensor likelihoods (1/2, 9/10, 1, 1/2). Event A contains the first two cells, giving $q(A) = 220/317 \approx 0.694$. Jev returns $p(A) = 0.84$ through Noul and 0.97 through Choice, with the same state, instructions, and event definitions. At $c = 0.10$, the 0.90 action threshold separates these answers: Noul leads to deferral with loss 0.10, whereas Choice leads to answering yes with expected loss $97/317 \approx 0.306$. A probability shift of 0.13 thus increases loss by about 0.206 in this example. The supplement preserves both exact requests, responses, and the selection rule.

H Decision Geometry

For 0–1 class loss and defer cost c , the Bayes risk and headroom below always defer are

$$L_c^*(q) = \min\{c, 1 - \|q\|_\infty\},$$

$$H_c(q) = \max\{0, \|q\|_\infty - (1 - c)\}.$$

For a normalized vector, the controller acts exactly when $\max_i p_i \geq 1 - c$. For posterior TV $\delta = \text{TV}(p, q)$, its regret to the oracle is at most 2δ ; an act/defer disagreement sharpens this to δ . If $|p_{(1)} - (1 - c)| > \text{TV}(p, p')$, two representations have the same act/defer status, and if $p_{(1)} - p_{(2)} > 2\text{TV}(p, p')$, they have the same maximizing class. Finally, aligning separate reports $x = \hat{P}(A)$ and $z = \hat{P}(A^c)$ gives

$$\text{TV}((1 - x, x), (z, 1 - z)) = |x + z - 1|,$$

so the complement residual lower-bounds the worse representation-specific posterior error by half its magnitude. These bounds use TV to evaluator truth when claiming regret; pairwise representation TV alone does not bound true downstream loss. To see the regret bound, let a_p, a_q minimize predicted and true expected loss. Insert and subtract their predicted losses. Predicted optimality makes the intervening difference non-positive, while each class-action loss changes by at most δ and defer loss does not change. This gives 2δ in general and δ if one selected action is defer. The margin claims follow from $|p_i - p'_i| \leq \text{TV}(p, p')$ for every

coordinate. The complement bound is the triangle inequality for the two aligned binary distributions and their common truth.

I Symmetric Post-Hoc Checks

For reference, the prespecified 15-bin sampled-outcome ECE values are 0.0691, 0.1691, 0.1176, and 0.1778 for Kev, base, posttrained, and Jev. These use one realized outcome per root and are distinct from the post-hoc exact-posterior summaries below. Applying the same specified validity threshold, relation tolerances, and lower-bound rule to all four complete configurations found that none met the broad conjunction (Table 8). This descriptive extension is useful context for the prospectively specified Jev evaluation: the benchmark exposes different contract failures across every configuration rather than a defect unique to one interface. The exact-posterior calibration column bins the reported top-class probability against the mean exact posterior mass of that selected class. It is distinct from sampled-outcome ECE and was not a prespecified primary endpoint.

Table 8: Symmetric all-configuration check. Exact- q gap is a 15-bin top-class calibration summary against evaluator posterior mass.

Configuration	Native validity	Exact- q gap	Broad contract
Kev typed	1.0000	0.0618	No
Matched-base likelihood	1.0000	0.1441	No
Posttrained likelihood	1.0000	0.0877	No
Jev-1.13.0	0.9946	0.1952	No

J Full Relation Estimates

Tables 9, 10, 11, and 12 report all ten relations, including those not displayed in the main comparison. Values summarize the existing primary relation records; this consolidated presentation adds no model calls. Product eligibility excludes zero conditioning mass. Missing or rejected constituent responses reduce valid support further. Intervals use the specified 5,000 root bootstrap samples and are pointwise. The supplied CSV contains the unrounded values and full denominator fields.

K Accounting Views and Stratum Estimates

Table 13 distinguishes the four accounting views; Table 14 gives canonical posterior TV for each generator–encoding stratum.

Table 9: All continuous relation residuals for Kev typed. Estimates and intervals preserve each relation’s paired-valid support.

Relation	Mean [95% interval]	Valid roots	Eligible roots
Complement	0.5055 [0.4971, 0.5138]	1000	1000
Event/Choice interface	0.0629 [0.0610, 0.0648]	1000	1000
Coarsening	0.3554 [0.3489, 0.3620]	1000	1000
Intersection bounds	0.1553 [0.1507, 0.1598]	1000	1000
Product	0.1328 [0.1300, 0.1356]	913	913
Exact repeat	0.0000 [0.0000, 0.0000]	1000	1000
Option permutation TV	0.1143 [0.1104, 0.1181]	1000	1000
Rendering TV	0.0891 [0.0859, 0.0923]	1000	1000
Opaque-key TV	0.0797 [0.0779, 0.0815]	1000	1000
Irrelevant-metadata TV	0.0636 [0.0614, 0.0660]	1000	1000

Table 10: All continuous relation residuals for Base likelihood. Estimates and intervals preserve each relation’s paired-valid support.

Relation	Mean [95% interval]	Valid roots	Eligible roots
Complement	0.1155 [0.1133, 0.1177]	1000	1000
Event/Choice interface	0.2138 [0.2111, 0.2165]	1000	1000
Coarsening	0.2451 [0.2420, 0.2481]	1000	1000
Intersection bounds	0.0814 [0.0807, 0.0821]	1000	1000
Product	0.2682 [0.2673, 0.2691]	913	913
Exact repeat	0.0000 [0.0000, 0.0000]	1000	1000
Option permutation TV	0.3224 [0.3150, 0.3294]	1000	1000
Rendering TV	0.1329 [0.1315, 0.1343]	1000	1000
Opaque-key TV	0.0541 [0.0532, 0.0551]	1000	1000
Irrelevant-metadata TV	0.0261 [0.0255, 0.0268]	1000	1000

L Service and Channel Diagnostics

Jev returned 11,849 valid responses, 62 rejected responses, and 2 unavailable outcomes across its 11,913 eligible main requests. All returned responses identified Jev-1.13.0. Native validity is 0.9946 and service availability is 0.9998. Rejected and unavailable requests are excluded from probability scores but take the defer action in intended-denominator loss. This accounting keeps failures in the evaluated pipeline. Because low-cost deferral can lower loss by reducing coverage, loss must be interpreted together with validity and deferral.

A post-hoc serialization sensitivity normalizes 57 rejected Choice vectors with finite, nonnegative entries and total-mass error no greater than 0.0100. Counting these as parseable would raise effective validity to 11,906/11,913 (0.9994), consistent with rounding near the simplex boundary. Primary responses remain rejected and unscored, and this sensitivity does not change failure of the joint benchmark criterion. Appendix E documents inclusive decimal-threshold handling so that a numerical boundary is not mistaken for a scientific effect.

The likelihood proxies have different normalization

behavior. Mean mass on the allowed answer tokens is 0.0110 for base (median 0.0099) and 0.6570 for post-trained (median 0.6932). A sharp distribution conditional on a selected answer set can thus discard most of the model’s full-vocabulary probability. Such a proxy should not be described as an unconditional semantic-event posterior.

A separate diagnostic asked posttrained to generate JSON probabilities. All 400 outputs satisfied the strict parser without repair. Relative to its likelihood channel, the probability gap was 0.1737 [0.1645, 0.1834] and the action-change rate was 0.5796 [0.5445, 0.6146]. This comparison tests two elicitation channels rather than a native probability head. Another diagnostic removed context: Kev and posttrained changed probabilities by 0.2964 [0.2875, 0.3052] and 0.2269 [0.2230, 0.2308] respectively on 200 roots and 400 paired cases each. Removing information is not an equivalence test, so those responses do not enter coherence claims.

Signed complement mass (post-hoc revision diagnostic). Recomputing $s = p(A) + p(\neg A)$ from the preserved fraction exports gives Table 15. Every Kev and posttrained pair has excess mass; the posttrained median is 1.8151, with range 1.5743–1.9150. These summaries distinguish systematic excess from cancellation

Table 11: All continuous relation residuals for Posttrained likelihood. Estimates and intervals preserve each relation’s paired-valid support.

Relation	Mean [95% interval]	Valid roots	Eligible roots
Complement	0.8000 [0.7982, 0.8017]	1000	1000
Event/Choice interface	0.1126 [0.1085, 0.1168]	1000	1000
Coarsening	0.2836 [0.2776, 0.2896]	1000	1000
Intersection bounds	0.0358 [0.0342, 0.0375]	1000	1000
Product	0.1406 [0.1389, 0.1424]	913	913
Exact repeat	0.0000 [0.0000, 0.0000]	1000	1000
Option permutation TV	0.3470 [0.3376, 0.3562]	1000	1000
Rendering TV	0.1026 [0.1002, 0.1050]	1000	1000
Opaque-key TV	0.2105 [0.2074, 0.2136]	1000	1000
Irrelevant-metadata TV	0.0767 [0.0750, 0.0783]	1000	1000

Table 12: All continuous relation residuals for Jev-1.13.0. Estimates and intervals preserve each relation’s paired-valid support.

Relation	Mean [95% interval]	Valid roots	Eligible roots
Complement	0.0929 [0.0890, 0.0967]	1000	1000
Event/Choice interface	0.1938 [0.1873, 0.2002]	1000	1000
Coarsening	0.1751 [0.1690, 0.1813]	990	1000
Intersection bounds	0.0200 [0.0173, 0.0227]	999	1000
Product	0.0836 [0.0804, 0.0868]	912	913
Exact repeat	0.0188 [0.0176, 0.0200]	1000	1000
Option permutation TV	0.0534 [0.0503, 0.0567]	976	1000
Rendering TV	0.0563 [0.0534, 0.0592]	977	1000
Opaque-key TV	0.0644 [0.0614, 0.0674]	976	1000
Irrelevant-metadata TV	0.0469 [0.0444, 0.0493]	979	1000

Table 13: Four accounting views retained for every configuration.

View	Question
Eligible	Was the semantic transformation defined?
Native valid	Did the returned object satisfy its interface contract?
Paired valid	Are all vectors needed for this contrast scoreable?
Intended pipeline	What loss results when failure triggers defer?

of high and low sums. They do not isolate a causal explanation: answer-label preference, event interpretation, arithmetic, and readout construction remain competing possibilities. No contextual recalibration or new label-order intervention was performed.

M Cross-Family Extension Details

M.1 Readout comparison on the shared subset

Table 16 evaluates the four new readouts on a shared subset. Both likelihood readouts return valid vectors throughout, yet complement gaps remain large: 0.6469 for Mistral and 0.2048 for OLMo. At $c = 0.10$, Mistral’s canonical loss is 0.1734; OLMo likelihood always defers. This extends the observation that validity alone does not ensure coherent reports or beneficial decisions.

Generated JSON has a different limitation: only 59 Mistral and 60 OLMo canonical outputs are valid. OLMo’s canonical mean excludes one empty stratum, and its complement mean covers five of eight strata. These are conditional summaries, not full-population accuracy estimates. On 59 paired-valid roots, Mistral likelihood-minus-JSN TV is 0.1145 [0.0454, 0.1855]; OLMo’s contrast is -0.0164 $[-0.0902, 0.0558]$ on 60 roots. Mistral JSON’s lower pipeline loss also accompanies greater deferral, including invalid outputs.

Event and Choice compile to identical model-facing inputs for these four readouts. Their zero gap is an

Table 14: Canonical posterior TV by prespecified generator–encoding stratum. Every specified stratum is shown, including reversals of aggregate ordering.

Stratum	Kev typed	Matched base	Posttrained	Jev
Finite population / joined tables	0.2696	0.4061	0.2381	0.2429
Finite population / nested records	0.1866	0.3905	0.3408	0.1858
Joint attributes / joined tables	0.1764	0.3097	0.2468	0.4017
Joint attributes / nested records	0.1868	0.2937	0.3489	0.3831
Missing rule / joined tables	0.4545	0.4111	0.3430	0.1931
Missing rule / nested records	0.1598	0.3614	0.3516	0.1272
Noisy sensor / joined tables	0.2002	0.4230	0.2871	0.2146
Noisy sensor / nested records	0.2457	0.3961	0.4054	0.3935
Eight-stratum macro	0.2350	0.3740	0.3202	0.2677

Table 15: Signed complement sums on 1,000 pairs per configuration. Post-hoc summaries computed during revision; all configurations retained.

Configuration	Mean	Minimum	Maximum	$s > 1$
Kev	1.5055	1.0485	1.8153	1000
Base	1.1103	0.9344	1.3733	861
Posttrained	1.8000	1.5743	1.9150	1000
Jev	1.0641	0.6700	1.3900	786

identity control, not an independent test of Jev’s native interface contrast. The tables below report all six contrasts, ten relations, costs, and support cells; the JSON–JSON canonical comparison has only 21 roots across four strata. The extension broadens the tested configurations, but does not add independent worlds or schema-constrained generation.

The extension evaluates two instruction-tuned checkpoints through two readouts each. Mistral-7B-Instruct-v0.3 uses revision `c170c708c41dac9275d15a8fff4eca08d52bab71`; OLMo-2-1124-7B-Instruct uses `470b1fba1ae01581f270116362ee4aa1b97f4c84`. Exact tokenizer and asset inventories, runtime receipts, and request records are retained in separate verified evidence backups. Collection used one NVIDIA A100-SXM4-80GB, Python 3.13.5, PyTorch 2.8.0 with CUDA 12.8, and Transformers 5.17.0. Both models ran in BF16 without quantization, CPU offload, or fitted calibration.

The main subset contains 200 historical roots, 25 in each generator–encoding stratum. Each readout has 2,400 intended request slots, of which 15 conditionals are not applicable; 2,385 requests remain. A separate 477-case development allocation per readout checked tokenization, runtime and timing and contributes no evaluation result. Two models, two readouts, and both

phases produced 11,448 terminal attempts, with zero retries. The likelihood readout normalizes only the admitted single-token answer IDs and retains the total vocabulary mass on those IDs. The generated readout uses greedy decoding with at most 128 new tokens and a strict JSON parser. Parse and simplex failures are rejected without extracting, normalizing, or repairing a vector.

For every new model/readout, Event *A*, binary Choice and exact repeat have byte-identical compiled prompts, token IDs, answer mappings and generation limits on all 200 roots. Their paired-valid vectors match exactly (200 pairs for each likelihood readout, 140 for Mistral JSON, 196 for OLMo JSON). Their zero residuals are therefore deterministic identity controls. The extension does not replicate Jev’s native Noul-versus-Choice intervention; complement, coarsening and the other nonidentical transformations remain distinct tests.

Table 17 separates parser outcomes. Sixteen rejected Mistral responses generated exactly 128 tokens; no OLMo response reached that count. Median generated lengths were 25 and 21 tokens, respectively. These descriptive diagnostics do not assign a causal mechanism to a rejection. Strict JSON reporting here is an unconstrained generated-text channel, not a provider-enforced typed schema.

Every main collection completed its intended allocation. Development plus main collection occupied 44.77 minutes for Mistral and 41.54 minutes for OLMo, including model loading; acquisition time is separate. Runtime records, collection source, and development/main archives are retained with the artifact.

The extension’s six unordered configuration contrasts were specified before these outputs. Probability-score contrasts use roots valid for both configurations; decision contrasts include all 200 roots, assigning rejected

Table 16: Four new configurations on the same 200-world subset. Validity uses all eligible requests; parentheses give valid canonical or paired-valid roots. Loss and defer rate retain all 200 roots at $c = 0.10$. Rejected outputs defer. Valid-only means average nonempty strata; full support and pointwise intervals are in Appendix M.

Configuration	Valid requests	Canonical TV (n)	Complement gap (n)	Loss	Defer
Mistral likelihood	2385/2385	0.4799 (200)	0.6469 (200)	0.1734	0.8900
Mistral JSON	1454/2385	0.3937 (59)	0.4848 (139)	0.1119	0.9650
OLMo likelihood	2385/2385	0.4261 (200)	0.2048 (200)	0.1000	1.0000
OLMo JSON	1248/2385	0.4220 (60)	0.2014 (103)	0.1142	0.9700

Table 17: Strict generated-JSON rejection reasons across all 2,385 main requests per model. Categories are mutually exclusive parser outcomes; none is repaired. These are protocol-validity failures, distinct from relation residuals on valid reports.

Reason	Mistral	OLMo
Invalid JSON	528	4
Wrong key support	0	536
Nonnumeric probability	21	0
Trailing content	8	0
Out-of-range probability	62	1
Simplex sum mismatch	312	596
Total rejected	931	1137

canonical outputs to defer. Marginal valid-only estimates average the nonempty strata equally. If a stratum has no valid outputs, it cannot contribute to that probability estimate; it remains present in intended-denominator decisions. The full support inventory accompanies each relation and contrast. Intervals are pointwise paired root-bootstrap summaries, not multiplicity-adjusted evidence of superiority. Historical subset results are contextual: prompts, precision, channel and collection time differ, so they do not identify an architectural effect or constitute a new-data replication.

Tables 18–23 report the complete subset estimates, intervals and selected support counts. The machine-readable independent support audit retains all relation-specific support, including zero-valid cells. OLMo JSON has no valid canonical output in the missing-rule joined-table stratum. Its complement estimate spans five strata, and its intersection and product estimates span only three. The paired JSON–JSON canonical comparison uses 21 roots across four strata. These valid-only quantities describe different conditional populations; they must not be read as eight-stratum full-pipeline rankings.

N Additional Related-Work Context

Readout and representation sensitivity are established: paraphrases, answer slots, context, and verbalized-versus-token readout can change confidence conclusions (Mahaut et al., 2024; Kim and Kang, 2026). Concurrent Jev studies cover task calibration, review routing, and factuality judging (Rafe and Das, 2026; Huang et al., 2026; Ibrahim and Zaki, 2026), option-name/rubric reassignment (Sun et al., 2026), request configuration and question/context ablations (Zhang et al., 2026; Guo et al., 2026), answer-preserving context additions (Xu, 2026), and semantic selections with downstream quantities (Deng et al., 2026). Belief-choice consistency and policy-dependent elicitation further motivate separating reported probabilities from decisions (Yamin et al., 2026; Kudum et al., 2026). To our knowledge, these studies do not jointly evaluate exact finite-world posteriors, validated event relations, cross-interface alignment, served failures, and paired decision regret. Our contribution is their combination in one benchmark, rather than any one test in isolation.

O Post-Hoc Certificates and Fixed Policies

This analysis reuses preserved predictions; it does not modify the primary collection or repair rejected outputs. Its specification was recorded after descriptive certificate ratios had been inspected. It retains every configuration, the five fixed costs, and 5,000 shared eight-stratum root-bootstrap replicates (seed 2026092408). All intervals are pointwise descriptive estimates. They quantify sampling over these worlds, not repeated API collection or generalization to model families. The evaluator reconstructs $q(A)$ from public counts and likelihoods and checks exact agreement with all aligned evaluator targets. All 1,000 Event/Choice, Event/repeat, and Event/complement pairs are complete for each configuration.

For a valid report p , the controller minimizes $(p, 1 - p, c)$ over (no, yes, defer), in that order at ties. The evaluator retains *all* minimizers under exact rational

Table 18: Canonical subset results with 95% pointwise root-bootstrap intervals. Probability scores use the stated valid support; expected decision loss uses all 200 roots at $c = 0.10$. Historical rows reuse earlier responses on exactly this subset and are not a controlled architecture comparison.

Configuration	n valid	TV	Expected Brier	Expected loss
Mistral likelihood	200	0.4799 [0.4517, 0.5099]	0.9696 [0.9275, 1.0132]	0.1734 [0.1531, 0.1942]
Mistral JSON	59	0.3937 [0.3506, 0.4358]	0.8126 [0.7548, 0.8725]	0.1119 [0.1001, 0.1271]
OLMo likelihood	200	0.4261 [0.3955, 0.4583]	0.8566 [0.8224, 0.8913]	0.1000 [0.1000, 0.1000]
OLMo JSON	60	0.4220 [0.3767, 0.4666]	0.8224 [0.7598, 0.8884]	0.1142 [0.1020, 0.1299]
Kev typed (historical)	200	0.2349 [0.2179, 0.2522]	0.6239 [0.6028, 0.6445]	0.1079 [0.1016, 0.1149]
Base likelihood (historical)	200	0.3785 [0.3601, 0.3967]	0.7474 [0.7250, 0.7698]	0.1000 [0.1000, 0.1000]
Post likelihood (historical)	200	0.3296 [0.3111, 0.3477]	0.7011 [0.6763, 0.7257]	0.1000 [0.1000, 0.1000]
Jev (historical)	200	0.2645 [0.2475, 0.2818]	0.6448 [0.6123, 0.6761]	0.1422 [0.1240, 0.1603]

Table 19: Mistral relation residuals with pointwise intervals. Each readout has 200 intended roots, except product (185 eligible). n counts complete valid constituents. Missing strata are excluded only from valid-only aggregates; the machine-readable support inventory identifies every exclusion.

Relation	n likelihood	Likelihood residual	n JSON	JSON residual
Event/Choice	200	0.0000 [0.0000, 0.0000]	140	0.0000 [0.0000, 0.0000]
Complement	200	0.6469 [0.6340, 0.6600]	139	0.4848 [0.4423, 0.5269]
Coarsening	200	0.2882 [0.2730, 0.3034]	51	0.2055 [0.1664, 0.2448]
Intersection bound	200	0.1091 [0.1052, 0.1131]	124	0.2184 [0.1904, 0.2490]
Product	185	0.1575 [0.1527, 0.1625]	146	0.3279 [0.2957, 0.3626]
Permutation	200	0.6167 [0.5891, 0.6432]	39	0.4039 [0.3559, 0.4553]
Rendering	200	0.1363 [0.1291, 0.1441]	31	0.1197 [0.0719, 0.1636]
Opaque keys	200	0.1347 [0.1276, 0.1420]	44	0.0772 [0.0414, 0.1209]
Irrelevant metadata	200	0.1398 [0.1324, 0.1473]	47	0.0895 [0.0486, 0.1329]
Exact repeat	200	0.0000 [0.0000, 0.0000]	140	0.0000 [0.0000, 0.0000]

q when deciding strict suboptimality. Different actions can both be Bayes-optimal; an action flip alone supplies no strictly positive universal regret bound. Policies receive only probability reports and the declared cost. Evaluator truth appears only in scoring.

For Eq. 1, sorting the two reports gives two cases: if q lies between them, the two distances sum to their separation; otherwise their sum is the separation plus twice the distance to the nearer endpoint. Expanding the squares around m proves Eq. 2. These checks concern average pair error, not canonical multiclass TV. Expected Bernoulli Brier adds $q(1-q)$; the two-coordinate categorical convention multiplies the complete identity by two.

The query costs are one for Event or Choice and two for averaging or agreement-deferral, including the corresponding repeat controls. Always defer needs no query. The oracle is evaluator-only and is not deployable. The comparison with average single-query loss means selecting an interface uniformly at random independently of the world. It is not the better interface chosen using gold. Averaging and two-report equality projection are the same policy, not separate methods. No full-event graph projection or fitted reconciliation was run.

Local exact repeats are identical, so repeat controls

reduce to Event-only. Jev repeats differ, but the preserved request order prevents a causal attribution of interface effects independent of service variation. Absolute expected loss is primary. Secondary headroom captured is $(c - \bar{L})/(c - \bar{L}^*)$, using the same binary cohort, with its denominator recomputed in each bootstrap. It can be negative and becomes unstable near zero; normalizing does not create available improvement. Full aggregates retain denominators, flags, exact ties, and per-root helps/harms rather than clipping negative ratios.

Figures 3 and 4 display the decomposition and primary-cost loss/coverage. Table 26 gives the full fixed cost grid. The secondary complement decomposition, complete policy intervals, coverage, regret, repeat controls, and helps/harms are supplied as machine-readable tables in the review artifact.

Table 20: OLMo relation residuals with pointwise intervals. Each readout has 200 intended roots, except product (185 eligible). n counts complete valid constituents. Missing strata are excluded only from valid-only aggregates; the machine-readable support inventory identifies every exclusion.

Relation	n likelihood	Likelihood residual	n JSON	JSON residual
Event/Choice	200	0.0000 [0.0000, 0.0000]	196	0.0000 [0.0000, 0.0000]
Complement	200	0.2048 [0.2012, 0.2084]	103	0.2014 [0.1742, 0.2297]
Coarsening	200	0.4655 [0.4599, 0.4711]	58	0.2456 [0.1977, 0.2937]
Intersection bound	200	0.0062 [0.0053, 0.0071]	46	0.2206 [0.1730, 0.2713]
Product	185	0.0984 [0.0958, 0.1010]	45	0.3613 [0.3201, 0.4026]
Permutation	200	0.4890 [0.4633, 0.5152]	34	0.4036 [0.3343, 0.4757]
Rendering	200	0.1278 [0.1231, 0.1328]	21	0.2084 [0.1540, 0.2684]
Opaque keys	200	0.1741 [0.1703, 0.1779]	40	0.0641 [0.0304, 0.1011]
Irrelevant metadata	200	0.0651 [0.0617, 0.0687]	44	0.0694 [0.0360, 0.1046]
Exact repeat	200	0.0000 [0.0000, 0.0000]	196	0.0000 [0.0000, 0.0000]

Table 21: Expected canonical decision loss for the complete fixed cost grid, all 200 roots per configuration. Rejected and unavailable outputs defer. Always-defer loss equals the column cost.

Configuration	$c = .02$	$c = .05$	$c = .10$	$c = .20$	$c = .40$
Mistral likelihood	0.0200	0.0581	0.1734	0.3124	0.5635
Mistral JSON	0.0200	0.0523	0.1119	0.2140	0.4304
OLMo likelihood	0.0200	0.0500	0.1000	0.2000	0.4304
OLMo JSON	0.0299	0.0640	0.1142	0.2177	0.4195
Kev typed (historical)	0.0200	0.0500	0.1079	0.1999	0.3913
Base likelihood (historical)	0.0200	0.0500	0.1000	0.2000	0.3844
Post likelihood (historical)	0.0200	0.0500	0.1000	0.2000	0.4012
Jev (historical)	0.0212	0.0784	0.1422	0.2568	0.4116

Table 22: All six planned configuration contrasts (left minus right). TV uses paired-valid canonical roots; loss uses all 200 roots at $c = 0.10$. Intervals are pointwise, without multiplicity-adjusted superiority claims. The full result file also includes Brier and the other fixed costs.

Contrast	n TV	TV difference	Expected-loss difference
Mistral likelihood – Mistral JSON	59	0.1145 [0.0454, 0.1855]	0.0615 [0.0342, 0.0871]
Mistral likelihood – OLMo likelihood	200	0.0538 [0.0331, 0.0750]	0.0734 [0.0531, 0.0942]
Mistral likelihood – OLMo JSON	60	0.0790 [0.0015, 0.1582]	0.0591 [0.0368, 0.0816]
Mistral JSON – OLMo likelihood	59	-0.0780 [-0.1505, -0.0077]	0.0119 [0.0001, 0.0271]
Mistral JSON – OLMo JSON	21	-0.1047 [-0.1629, -0.0506]	-0.0024 [-0.0226, 0.0179]
OLMo likelihood – OLMo JSON	60	-0.0164 [-0.0902, 0.0558]	-0.0142 [-0.0299, -0.0020]

Table 23: Valid-root support by the eight fixed strata (25 intended roots each): finite population (F), joint attributes (J), missing rule (M), and noisy sensor (N), each in joined-table (T) and nested-record (R) form. Zero-support cells are excluded from valid-only means; all cells remain in decision loss. Canonical and Event/Choice supports need not coincide.

Configuration and estimand	F/T	F/R	J/T	J/R	M/T	M/R	N/T	N/R
Mistral likelihood, canonical	25	25	25	25	25	25	25	25
Mistral likelihood, Event/Choice	25	25	25	25	25	25	25	25
Mistral JSON, canonical	8	1	8	2	15	16	8	1
Mistral JSON, Event/Choice	25	12	18	7	24	24	19	11
OLMo likelihood, canonical	25	25	25	25	25	25	25	25
OLMo likelihood, Event/Choice	25	25	25	25	25	25	25	25
OLMo JSON, canonical	2	6	6	9	0	21	8	8
OLMo JSON, Event/Choice	25	22	25	25	25	24	25	25

Table 24: Post-hoc absolute-error decomposition for Event/Choice: mean and pointwise 95% stratified root-bootstrap intervals. Certificate plus uncertified error equals pair absolute error before rounding.

Configuration	Pair absolute error	Observable certificate	Uncertified error
Kev	0.29397 [0.28537, 0.30287]	0.03143 [0.03048, 0.03240]	0.26254 [0.25384, 0.27140]
Base	0.25692 [0.24884, 0.26514]	0.10689 [0.10553, 0.10816]	0.15004 [0.14149, 0.15864]
Posttrained	0.37758 [0.36562, 0.38971]	0.05632 [0.05423, 0.05836]	0.32126 [0.30941, 0.33318]
Jev	0.18567 [0.18023, 0.19105]	0.09688 [0.09362, 0.10010]	0.08879 [0.08376, 0.09393]

Table 25: Post-hoc primary reporting contrasts at $c = 0.10$: expected loss difference with pointwise 95% intervals. Negative favors the first policy. The average-singles comparator randomizes equally between Event and Choice. No multiplicity-adjusted significance claim is made.

Configuration	Mean minus average singles	Agreement minus mean
Kev	-0.00959 [-0.01347, -0.00581]	-0.01095 [-0.01503, -0.00733]
Base	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]
Posttrained	-0.05297 [-0.05964, -0.04634]	-0.01816 [-0.02399, -0.01291]
Jev	-0.03463 [-0.03886, -0.03049]	-0.00236 [-0.00411, -0.00077]

Table 26: Post-hoc binary expected loss across all fixed costs. Every configuration has 1,000 intended roots and complete inputs. Event/Choice use one query; mean/agreement and repeat controls use two. Always defer has loss c . Oracle is evaluator-only. Full pointwise intervals and coverage remain in the machine-readable results.

Configuration	c	Event	Choice	Mean	Agreement	Repeat mean	Repeat agreement	Oracle
Kev	0.02	0.0200	0.0200	0.0200	0.0200	0.0200	0.0200	0.0165
Kev	0.05	0.0512	0.0516	0.0506	0.0501	0.0512	0.0512	0.0411
Kev	0.10	0.1669	0.1579	0.1528	0.1419	0.1669	0.1669	0.0813
Kev	0.20	0.3128	0.3052	0.3122	0.2885	0.3128	0.3128	0.1571
Kev	0.40	0.3988	0.3986	0.3967	0.3950	0.3988	0.3988	0.2689
Base	0.02	0.0200	0.0200	0.0200	0.0200	0.0200	0.0200	0.0165
Base	0.05	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0411
Base	0.10	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.0813
Base	0.20	0.2000	0.2009	0.2000	0.2000	0.2000	0.2000	0.1571
Base	0.40	0.4400	0.4899	0.4667	0.4400	0.4400	0.4400	0.2689
Posttrained	0.02	0.0200	0.0680	0.0200	0.0200	0.0200	0.0200	0.0165
Posttrained	0.05	0.0500	0.1146	0.0792	0.0500	0.0500	0.0500	0.0411
Posttrained	0.10	0.2545	0.2326	0.1906	0.1724	0.2545	0.2545	0.0813
Posttrained	0.20	0.4503	0.3388	0.3693	0.3236	0.4503	0.4503	0.1571
Posttrained	0.40	0.4899	0.4830	0.4899	0.4828	0.4899	0.4899	0.2689
Jev	0.02	0.0200	0.0292	0.0200	0.0200	0.0200	0.0200	0.0165
Jev	0.05	0.0502	0.0975	0.0497	0.0501	0.0501	0.0501	0.0411
Jev	0.10	0.0965	0.1705	0.0989	0.0965	0.0964	0.0964	0.0813
Jev	0.20	0.1853	0.2696	0.2028	0.1853	0.1844	0.1858	0.1571
Jev	0.40	0.3109	0.3514	0.3302	0.3166	0.3093	0.3099	0.2689

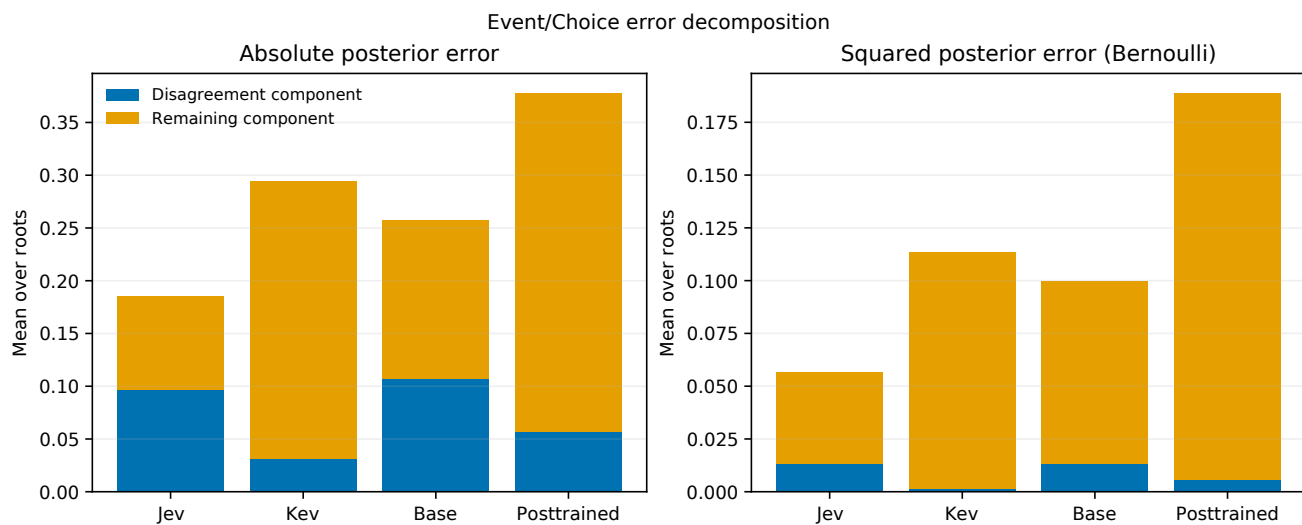


Figure 3: Post-hoc Event/Choice error decomposition on all 1,000 pairs per configuration. The observable disagreement certificate and the remaining component sum to the corresponding mean pair error. Ratios describe these reports and worlds; they are not causal attributions or a universal calibration bound.

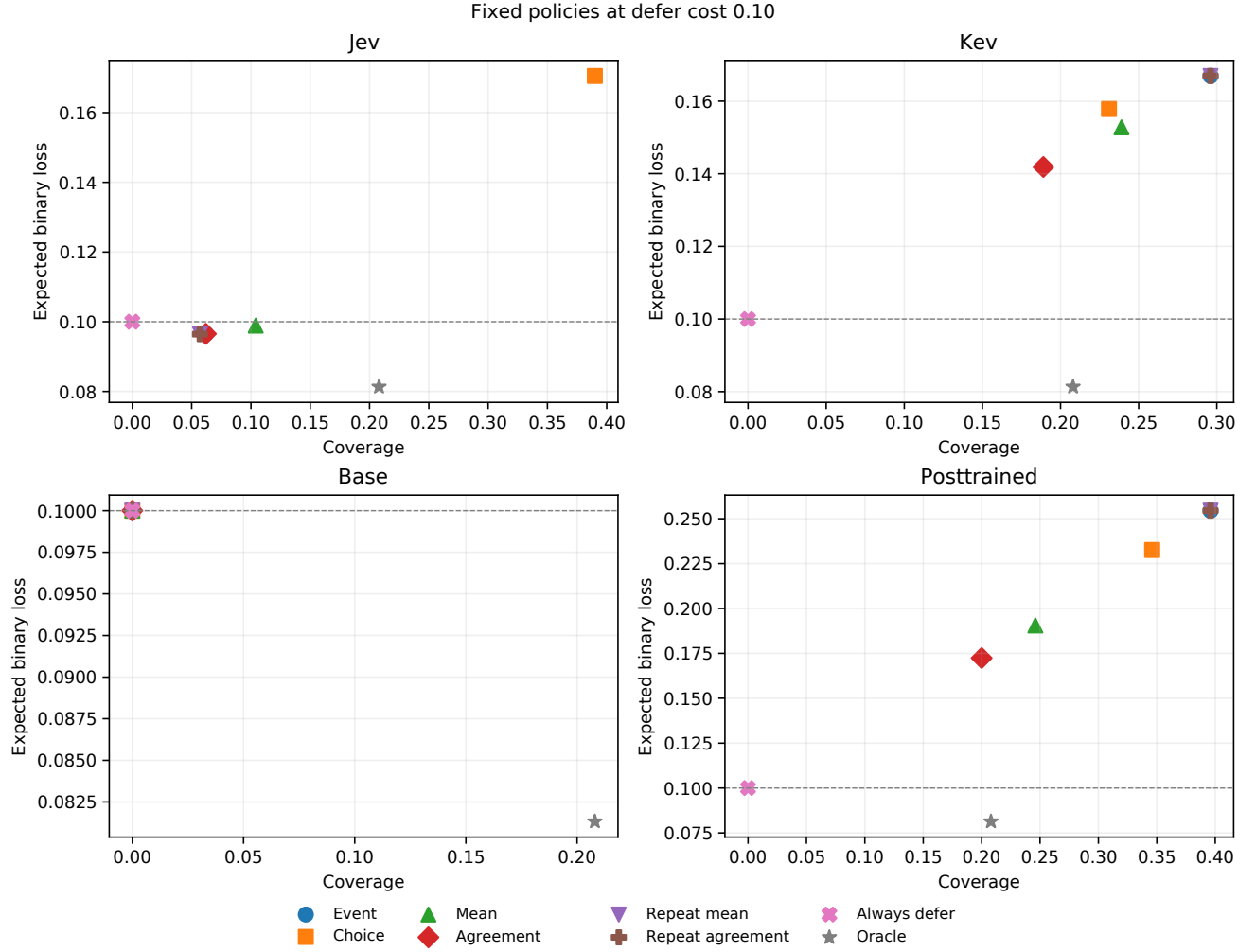


Figure 4: Post-hoc fixed-policy loss and class-action coverage at the primary defer cost $c = 0.10$. Each configuration is shown separately. Always defer is a cost reference; the oracle uses evaluator truth. The policy/contrast tables supply pointwise paired intervals.

P Averaging Across Action Regions

This elementary characterization was added during revision to explain the fixed-policy results. It changes no collected output, policy, cost, or primary estimand. Let $0 < c < 1/2$, with losses $L_c(0; q) = q$, $L_c(d; q) = c$, and $L_c(1; q) = 1 - q$. The report-based controller selects 0 on $[0, c]$, d on $(c, 1 - c)$, and 1 on $[1 - c, 1]$, so exact ties prefer a class over deferral.

Proof of Proposition 1. Each action region is convex. If both reports select the same action, their midpoint is in that region and $\Delta_c = 0$. For two adjacent regions, the midpoint selects one of their two actions. Substituting its true loss into Equation 3 gives half the difference between the selected and unselected losses. For opposite class actions, the randomized baseline has loss $\{q + (1 - q)\}/2 = 1/2$; subtracting it from each possible midpoint action loss gives the remaining rows of Table 27. These cases exhaust the action pairs. In particular, an opposite-class pair resolved to defer has penalty $c - 1/2 < 0$. The posterior q is used only to score the action, never by the policy.

Table 27: Exact per-world penalty of the mean policy versus uniformly selecting one of the two reports. The first column is an unordered action pair; rows apply when the stated midpoint action is feasible.

Original actions	Midpoint action	Δ_c
Same action	Same action	0
No, defer	No	$(q - c)/2$
No, defer	Defer	$(c - q)/2$
Defer, yes	Defer	$(c + q - 1)/2$
Defer, yes	Yes	$(1 - q - c)/2$
No, yes	No	$q - 1/2$
No, yes	Defer	$c - 1/2$
No, yes	Yes	$1/2 - q$

An analytic high-accuracy example. The observed same-baseline penalties occur in weak configurations. That empirical limitation is not a mathematical restriction on averaging. Fix any $c \in (0, 1/2)$ and choose

$$0 < \epsilon < \min\{c, (1 - 2c)/3\}.$$

Consider two equally weighted worlds. In the first, $q_1 = c - \epsilon$, $p_1 = c - \epsilon$, and $p'_1 = c + 3\epsilon$. The reports select no and defer; their midpoint $c + \epsilon$ defers. In the second, $q_2 = p_2 = p'_2 = \epsilon$, and every policy selects no. Every report is within 4ϵ of its true posterior. Averaging has population decision loss $(c + \epsilon)/2$, compared with $c/2 + \epsilon/4$ for randomized singles. Both are strictly better than always defer, yet averaging is worse by $\epsilon/4$. Meanwhile, averaging reduces the population expected one-coordinate Brier score by $2\epsilon^2$. Letting ϵ approach zero makes the reports arbitrarily accurate; the decision penalty also vanishes, consistent with posterior-error regret bounds. This is a constructed mathematical example, not an additional model run or evidence of performance in a high-accuracy regime.